

OVERALL CONCEPT · 总体概念

Human-Robot Labor Confluence · 人机合辙
'Zhe' is the imprint a wheel leaves on the rail. CONFLUENT TRACKS translates the century-old collaborative spirit of Jing-Zhang switchmen — 'guarding the switch with their lives' — into a spatial institution of human-robot relay at the physical labor layer: robots and drones take on hazardous work such as power-line inspection, utility-culvert inspection and fire reconnaissance in authorized pilots, while humans retain a three-stage chain of control — authorization, halt and acceptance; AI algorithms assist testing of urban traffic green-wave coordination and transit priority.
Long-Life Frame + Fast AI Infill
'Replace the model, not the city' : a 50-year long-life spatial frame plus fast modular AI infill slots, exploring the lifecycle mismatch between rapidly iterating AI models and long-lived urban physical structures.
50-Year Frame / 6-Month Iteration
The spatial frame carries a 50-year service life while AI modules iterate on a ~6-month cadence; handover stations, twin loops and the land-use grid reserve unified physical interfaces for successive algorithm generations.

THREE LAWS OF CONFLUENCE · 合辙三律

I · Physical Labor Collaboration
Robots are proposed to take on elevated, confined and toxic operations in authorized test scenarios, replacing front-line workers' hazardous exposure.
II · Human Sovereign Final Review
Human pre-authorization, a retained mid-task physical e-stop, and post-task expert acceptance sign-off — the three-stage chain of control is never ceded.
III · Pedestrian Priority in Public Space
Low-speed robots and UAVs in mixed spaces take pedestrian and vulnerable-group priority as a proposed test objective.

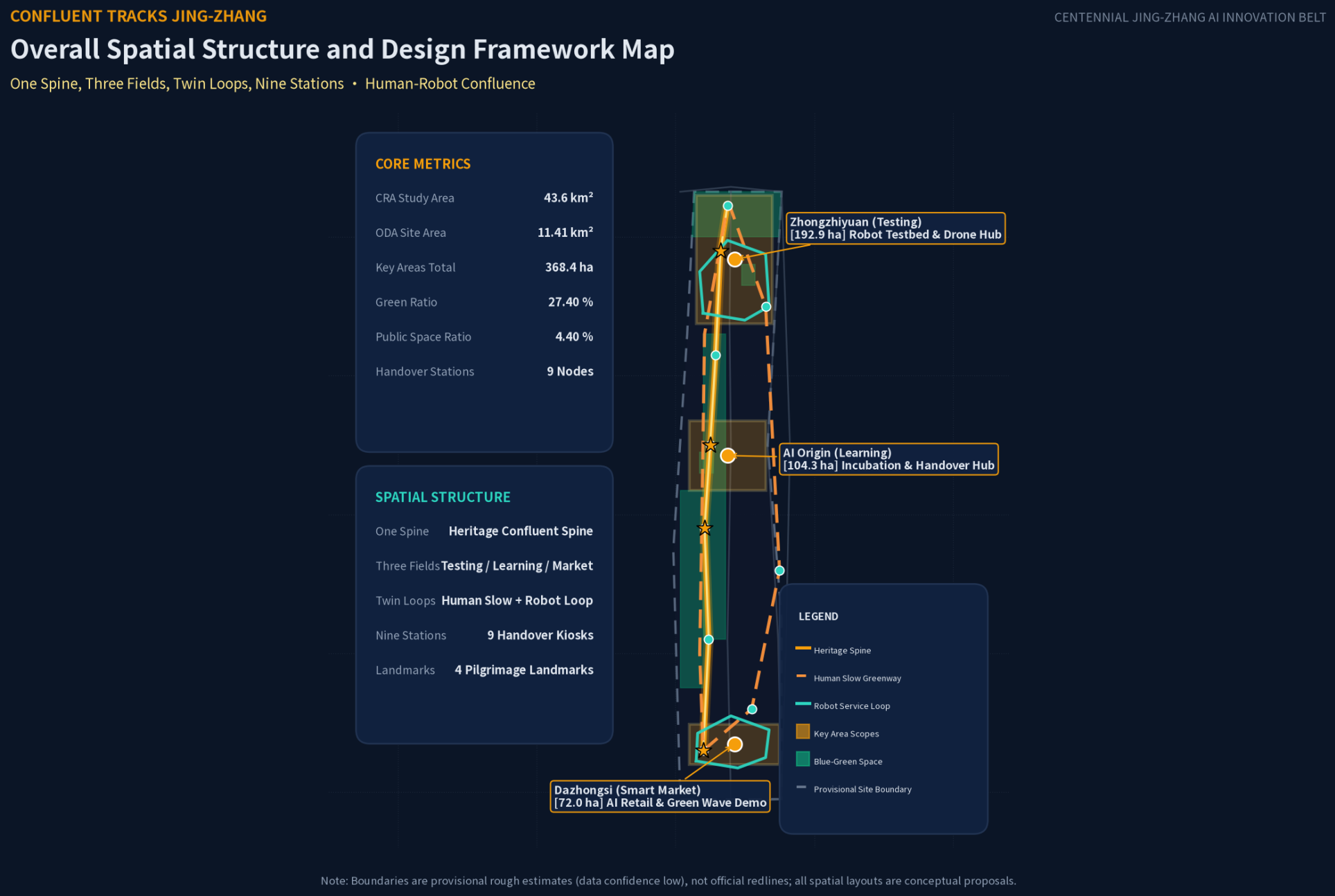
Regional Synergy (Conceptual Research)

Haidian original innovation — Future Science City (energy AI) / Huairou Science City (fundamental compute) / Beijing E-Town (embodied manufacturing) regional transfer — Jing-Jin-Ji (Tianjin Port, Xiong'an, Shijiazhuang, Baoding) demonstration and compute support. This chain is a hypothesis for validation, not an existing collaboration, approved regional plan or funding arrangement.

NAMING & IDENTITY · 命名与品牌

合辙京张 / CONFLUENT TRACKS JING-ZHANG
Subtitles: Zhongzhiyuan 'Confluent Testing Arena' · AI Origin Community 'Confluent Learning Garden' · Dazhongsi 'Confluent Smart Market' ; two wings: Zhongguancun Technology Services Wing · Xiaoyue River Scenario Enablement Wing.
The logo direction is modeled on Zhan Tianyou's herringbone switch: two parallel arrows in humanist orange (workers) and robot cyan (agents) within a railway circle; purely geometric, with no real trademarks or licensed fonts. International theme: 'From Switchman to Co-Pilot' .







MNR 2023 LAND-USE RECALCULATION

MNR code	Category	Area (ha)	Share
07	Residential	42.03	3.7 %
08	Public admin & services	182.13	16.0 %
0803	Cultural (direct code; child of 08)	78.08	6.8 %
09	Commercial services	259.13	22.7 %
1207	Urban & rural roads	367.28	32.2 %
14	Green & open space	201.77	17.7 %
1401	Park green space (direct code; child of 14)	10.85	1.0 %

The direct-code table uses exact geometry codes: 14=201.77 ha and 1401=10.85 ha; parent and child rows are separate and never double-counted. The dissolved green_space overlay is independently 312.70 ha / 27.40%; the stale 12.4 ha draft value is withdrawn.

URBAN RENEWAL FRAMEWORK · 更新框架

Low-Disruption Organic Renewal

Key renewal targets are Zhongzhiyuan potential land, near-campus blocks of the Origin Community, and the Dazhongsi commercial environment; AI agglomeration targets and industrial spatial scales are conceptual recommendations, and the total planned building volume remains pending confirmation without an official regulatory plan.

Retain-Renovate-Demolish · 拆改留

Retain heritage elements such as the Qinghuayuan Station historic site and quality existing buildings; renovate inefficient near-campus blocks and commercial environments; demolish only dilapidated structures confirmed by professional teams and valueless temporary buildings; build new mainly on potential land.

Conceptual Volumes · 概念体量

54 conceptual building volumes with a combined footprint of $\approx 3,710,756.5\text{ m}^2$ — marked as conceptual-volume footprints, neither surveyed existing conditions nor approved scales.

DEVELOPMENT INTENSITY · 开发强度

The unknown / null Principle

Indicators depending on undisclosed official controls — floor area ratio, building height and building coverage — keep status=unknown, value=null, with recalculation paths documented in assumptions.json and metrics.json; conceptual volumes never masquerade as statutory control values.

Spatial Supply & Talent Life

R&D land sits close to rail transit and the walking/cycling loop; commercial-service land connects to robot service loops; residential land is placed in quieter western and southern areas; green and public spaces are evenly distributed to support innovation and exchange.

Architectural Character

Key areas balance future-oriented and cultural expression; controls on height, volume and roof form are recorded as conceptual guidance pending official conditions — heritage humility, new-area futurism and transition-zone coordination.

CONFLUENT TRACKS JING-ZHANG

Spatial Diagrams and Design Moves for Three Key Areas

Each area shows a simplified polygon, nodes, mobility loops and a risk marker

Zhongzhiyuan • Arena

192.9 ha • North 5th Ring

[Role]

Robot testbed / user AI lab service hub

[Landmark]

Watchtower (stone wall + back)

[Risk]

Airspace safety and UAV runner pending

AI Origin • Learning Garden

104.3 ha • Wudaokou Core

[Role]

AI incubation / on-governance demonstration

[Landmark]

Handover Hub (infrastructure hub)

[Risk]

Heritage boundary and crossing pending

Dazhongsi • Smart Market

72.0 ha • Zhichunlu to Kailashan

[Role]

AI-native retail / modern services

[Landmark]

Green Wave Crossing (four-quadrant walk)

[Risk]

Cyclist and delivery robot right-of-way

Note: Key area boundaries are provisional; nodes, mobility and risks are conceptual diagrams, not official redlines or engineering alignments.

Detailed-design index of the three key areas

Unified Handover Interface

Before a task, authorization is confirmed by card or QR code; during a task it can be halted at any time in the halt zone; after a task, humans verify results in the acceptance zone. All three polygons are provisional and the designs are conceptual proposals for professional deepening.

Zhongzhiyuan • Confluent Testing Arena

≈ 192.9 ha (announcement reference 192.1 ha, comparison only) • North 5th Ring- Qing River

[Role] testing & validation field for inspection robots; proposed AI innovation testbed

[Landmark] Watchtower (drone nest + public viewing platform)

[Key projects] power-line / utility-culvert / bridge test grounds; Qing River waterfront garden

[Mobility] Fifth Ring Road area integration (conceptual, no redline conclusion)

[Risk] airspace and municipal authorizations pending; controls unknown

AI Origin Community • Confluent Learning Garden

≈ 104.3 ha • Tsinghua East Rd-Wudaokou

[Role] human-robot R&D and career-transition academy

[Landmark] Human-Robot Handover Hub (glazed visual authorization hall, S5 Wudaokou)

[Key projects] sci-tech incubation & transformation zones, healing gardens, launch functions

[Mobility] Wudaokou / Qinghua East Rd West station integration, campus-park slow links

[Risk] heritage boundary of Qinghuayuan Station site pending verification

Dazhongsi • Confluent Smart Market

≈ 72.0 ha • Zhichunlu-Xizhimen North

[Role] AI-native consumption and business cluster

[Landmark] Green-Wave Intersection (four-quadrant walking connection)

[Key projects] agents / smart terminals / content formats; delivery-robot narrow-lane microcirculation

[Mobility] metro linkage & greenway stitching; static traffic completed

[Risk] courier vs delivery-robot right-of-way requires field testing

Beiwei Community • Living Interface (Conceptual Research)

The residential living layer of the AI Origin Community: a community-level micro handover station (Type-C variant), an AI Living Lab unit and a community data dashboard; smart service stations, AI science corners, elderly-greeting and guide-companion routes. All content is conceptual exploration, not an existing collaboration or approved renewal.

12 Scenario Cards (3 Industrial Validation)

#	AI Scenario Card	Location	SWB Fallback
1	Power-Line Inspection Robot Partner (validation)	Zhongzhiyuan substation & line corridor	manual verification & breaker pull
2	Utility-Culvert Inspection Drone (validation)	Zhongzhiyuan utility tunnel	manual ventilation & manual entry
3	Traffic Green-Wave Coordination AI (validation)	Dazhongsi four-quadrant intersection	fixed timing + on-site police control
4	Bridge Structural Health Inspection	Qing River bridges & overpasses	manual basket inspection
5	Fire-Reconnaissance Vanguard Drone	belt-wide low-altitude grid corridors	traditional scout team entry
6	Night Quiet Cleaning Robot	twin loops & heritage park greenways	daytime manual sanitation
7	Guide-Cane Companion Robot	heritage park barrier-free greenway	white cane + volunteer guide
8	Commercial Micro-Delivery Robot	Dazhongsi dedicated narrow lanes	courier direct delivery
9	Emergency Medical AED Drone	belt-wide emergency low-altitude corridor	authorized ground-ambulance backup
10	Construction Safety AI Supervision	key-area renewal sites	conventional manual inspection
11	Autonomous Feeder Shuttle	station-integrated feeder routes	standard human-driven feeder bus
12	Solitary-Elderly Greeting Robot	Origin Community & senior zones	routine social-worker visits

Each card specifies data categories, anonymization, retention, access, human review, physical e-stop and complaint handling; response times, hotlines and access channels are proposed SLAs / test targets pending authorization — not current service commitments.

Three Spatial Conflict Resolution Prototypes

① Intersection Crosswalk Confluence A 1.5 m dedicated robot deceleration island and dynamic ground projection alerts; pedestrians hold the highest priority; robots decelerate to 2 km/h at 2 m and yield statically — validated against mixed-crossing collision-risk reduction.
② Courier vs Delivery-Robot Lane Separation A 1.2 m delivery-robot lane in Dazhongsi with a 15 cm sloped landscaped curb as physical buffering — tested against sideswipe-risk reduction between 25 km/h courier scooters and 6 km/h delivery robots.
③ Tactile Blind-Path Buffer A continuous 60 cm physical buffer on both sides of tactile paving along the 8.7 km greenway, barring charging kiosks or parked equipment; guide-companion robots hold a 50 cm lateral offset via UWB geofencing.

User Personas · 用户画像

Persona	Core need
Electrician Mr. Zhang (45)	fewer high-voltage climbs; robot partners take hazards
Traffic assistant Ms. Li	smarter signals, but she can take over
Engineer Xiao Wang (25)	a walkable ‘third space’
Heritage docent Mr. Chen	heritage respected; tactile, readable narrative
Visually impaired Ms. Liu	tactile & audible guidance; unobstructed blind paths
Courier Xiao Zhou	safe cycling space not crowded out by robots

AI Pilgrimage Landmarks · 朝圣地标

4 Landmarks (all conceptual) ① Centennial Switch Square (south Zhongzhiyuan) — switchman sculpture beside an inspection robot with a tactile relief timeline; ② Human-Robot Handover Hub (S5 Wudaokou) — visual authorization hall; ③ Watchtower (Qing River) — drone nest + public viewing platform; ④ Green-Wave Intersection (Dazhongsi) — data-flow lighting visualizes signal coordination at night.





Implementation Phasing · 实施分期

Phase	Period	Area	Focus
Near-term pilots	2026-2028	≈ 159.4 ha	three station pilots + green-wave segment
Mid-term network	2028-2030	≈ 297.4 ha	twin loops connected; nine stations formed
Long-term confluence	2030-2035	≈ 392.3 ha	43.6 km ² synergy & long-term operation

All phasing is a conceptual recommendation, not a development-timeline commitment or government approval.

Metrics & Evidence · 指标体系

Metric	Status	Value
site_area_sqm	known	11,412,825.4 m ²
green_ratio	known	27.4 % (3,127,039.4 m ²)
public_space_ratio	known	4.4 % (502,225.7 m ²)
road_network_length_m	known	50,618.2 m
slow_service_network_length_m	known	16,898.8 m
buildings / footprint	known	54 bldgs / 3,710,756.5 m ²
floor_area_ratio	unknown	null (pending official plan)
building_height_m_max	unknown	null (pending official plan)
building_density_official	unknown	null (pending official plan)

The compliance matrix covers Announcement 1.3/1.4/1.5 and agent.1-agent.6; 9 professional standards (all 5 mandatory addressed); all 15 design-depth items complete; all four self-check gates PASS.

Policy & Operation · 政策与运营

On-Duty Voucher & Access Timetable Hazardous inspection robots and autonomous shuttles entering public space pass closed-track validation and receive electronic on-duty vouchers before authorization; nighttime low-noise (22:00-06:00) and daytime off-peak schedules may be tested. All policy and funding arrangements are conceptual recommendations.
Long-Term Operation & Annual Congress An annual Human-Robot Confluence Congress, a developer challenge on hazardous-operation simulation tracks, a scenario-access channel for enterprises, public experience routes and conversion monitoring — under the international theme ‘From Switchman to Co-Pilot’ .
Three Risk Categories · 风险 ① Official redlines and key-area polygons not yet obtained — all spatial conclusions are provisional with a package-wide recalculation trigger; ② regulatory controls, road redlines, ownership and municipal pipelines are not public — related metrics remain unknown; ③ heritage scopes and facts (Qinghuayuan Station site, Juesheng Temple) await verification by cultural-relic authorities.